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University of Information Technology & Sciences

Test Case Report

Course : **Software Engineering and System Analysis Lab**

Course Code : **CSE0613226**

Submitted To:
Shad Ahmed
Lecturer
Md Faysal
Lecturer
Department Of CSE

Submitted By:
Md Nayem Islam
ID No: 04325105101041
Md Zayed Adnan Zabin
ID No: 04325105101034
Mahjabin Afroze Mahira
ID No: 04325105101007
Kanis Akter
ID No: 04325105101018

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Software Testing and Quality Assurance

Test Case Report

Project: Online Crime Reporting and Case Management System

Mapped CO: CO3

Tools: PHP (XAMPP), MySQL, PHPUnit for unit tests, browser-based functional testing

1. Objective

To design and execute a testing strategy for the Online Crime Reporting and Case Management System, covering unit tests for the core helper functions and integration tests for the key end-to-end flows of the three roles (Citizen, Police Officer, Administrator), including the case lifecycle (Pending, Verified, Investigating, Resolved, Closed), and to record the defects found and fixed.

2. Test Scenario

Verify that citizens can register, log in, submit crime reports with evidence and track them; that police officers can verify or reject reports and manage investigations; that administrators can manage users, officers and crime categories; and that every page blocks invalid input and unauthorised access with an appropriate message. Reports can also be searched by Case ID, keyword, status, category and date, and printed from the details pages.

3. Test Environment

[1] Web server and database: XAMPP (Apache, PHP, MySQL on port 3306 as set in config/database.php) with the database crime_reporting_system created from database.sql, including the case_status_history table

[2] Application: online_crime_reporting (PHP with PDO, session-based authentication, role-based access control, Case ID format CR-YYYY-NNNNN, Closed case status, status history)

[3] Seed data: administrator admin@crms.local, 8 crime categories, 6 help resources

[4] Charts: dashboards load Chart.js from a CDN, so an internet connection is needed to see the charts

[5] Test accounts: one citizen, a second citizen (for ownership checks) and one police officer

4. Test Strategy

Unit tests follow the Arrange, Act, Assert pattern. Test data was chosen using the techniques below.

[1] Equivalence partitioning: valid and invalid classes for each input, for example allowed and disallowed file types, valid and invalid email formats.

[2] Boundary value analysis: values at and just beyond each limit, for example name length 2 and 3, password length 7 and 8, title length 4, 5, 150 and 151, description length 19 and 20, file size 5,242,880 and 5,242,881 bytes, 5 and 6 evidence files, incident date today and tomorrow.

[3] Negative and security testing: SQL injection, script tags, invalid CSRF token, role escalation, disguised file uploads and access to another user's data.

[4] Integration tests follow complete flows: register, login, submit report, verify report, investigate, resolve, and notifications at each step.

5. Unit Test Cases

Core functions in includes/functions.php and includes/auth.php.

sanitize_input()

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
UT_01	Normal text	None	Input: "Adnan"	"Adnan"	"Adnan"	**Pass**
UT_02	Leading and trailing spaces	None	Input: " Theft report "	"Theft report"	"Theft report"	**Pass**
UT_03	HTML / script tag	None	Input: <script>alert(1)</script>	Tags escaped: <script>alert(1)</script>	Tags escaped: <script>alert(1)</script>	**Pass**
UT_04	Quote characters	None	Input: O'Brien "Jr"	O'Brien "Jr"	O'Brien "Jr"	**Pass**
UT_05	Backslash in input	None	Input: C:\Users\evidence	Backslashes preserved: C:\Users\evidence	Returned C:Usersevidence (backslashes removed by stripslashes)	**Fail**
UT_06	Empty string	None	Input: ""	""	""	**Pass**

validate_file_upload()

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
UT_07	Valid JPG file	Real JPEG file of 100 KB	name = photo.jpg, size = 102400, error = 0	success = true	success = true	**Pass**
UT_08	Valid PNG file	Real PNG file of 100 KB	name = scene.png, size = 102400, error = 0	success = true	success = true	**Pass**
UT_09	Valid PDF file	Real PDF file of 100 KB	name = statement.pdf, size = 102400, error = 0	success = true	success = true	**Pass**
UT_10	Upper-case extension	Real JPEG file	name = PHOTO.JPG	success = true	success = true (extension is lower-cased before checking)	**Pass**
UT_11	Size at upper boundary	Real PDF file	size = 5,242,880 bytes (exactly 5 MB)	Accepted	Accepted	**Pass**
UT_12	Size just above boundary	Real PDF file	size = 5,242,881 bytes	Rejected: "File size must not exceed 5 MB."	Rejected: "File size must not exceed 5 MB."	**Pass**
UT_13	Zero-byte file	Empty file named empty.pdf	size = 0	Rejected	Rejected: "The file content does not match its extension."	**Pass**
UT_14	Disallowed extension	Any file	name = virus.exe	Rejected: allowed types message	Rejected: "Only JPG, JPEG, PNG, PDF files are allowed."	**Pass**
UT_15	Script disguised as image	Text file containing PHP code	name = shell.jpg	Rejected: content does not match extension	Rejected: "The file content does not match its extension."	**Pass**
UT_16	Wrong extension for real content	Real JPEG file	name = photo.png	Rejected: content does not match extension	Rejected: "The file content does not match its extension."	**Pass**
UT_17	File with no extension	Any file	name = evidence	Rejected	Rejected: allowed types message	**Pass**
UT_18	No file selected	None	error = UPLOAD_ERR_NO_FILE	Rejected: "No file was uploaded."	Rejected: "No file was uploaded."	**Pass**

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
UT_19	Server size limit exceeded	None	error = UPLOAD_ERR_INI_SIZE	Rejected: "File exceeds the maximum allowed size."	Rejected: "File exceeds the maximum allowed size."	**Pass**
UT_20	Malformed upload array	None	error = array(0, 0)	Rejected: "Invalid file upload."	Rejected: "Invalid file upload."	**Pass**

generate_unique_filename()

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
UT_21	Filename format	None	Input: "Photo.JPG"	Matches 32 hex characters + _ + timestamp + .jpg	Matches ^[a-f0-9]{32}_[0-9]+\.jpg$	**Pass**
UT_22	Uniqueness	None	Call twice with the same name	Two different filenames	Two different filenames	**Pass**
UT_23	Path traversal in original name	None	Input: "../../evil.php.png"	Only the extension is reused; no path characters	Result ends with .png and contains no ../	**Pass**

get_status_badge_class()

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
UT_24	Each defined status	None	Pending, Verified, Rejected, Investigating, Resolved	bg-warning text-dark, bg-info text-dark, bg-danger, bg-primary, bg-success	Same classes returned	**Pass**
UT_25	Case-insensitive match	None	Input: "RESOLVED"	bg-success	bg-success	**Pass**
UT_26	Closed status (no dedicated colour)	None	Input: "Closed"	bg-secondary (default)	bg-secondary	**Pass**
UT_27	NULL status	None	Input: null	bg-secondary	bg-secondary	**Pass**

CSRF token functions

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
UT_28	Token generation	Fresh session	Call generate_csrf_token()	64-character hexadecimal string	64-character hexadecimal string	**Pass**
UT_29	Token reused in same session	Token already generated	Call generate_csrf_token() twice	Same token returned both times	Same token returned both times	**Pass**
UT_30	Correct token	Token generated	verify_csrf_token(<session token>)	true	true	**Pass**
UT_31	Wrong token	Token generated	verify_csrf_token("abc123")	false	false	**Pass**
UT_32	Empty token / no session token	No token in session	verify_csrf_token("")	false	false	**Pass**

format_date()

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
UT_33	Valid datetime	None	Input: "2026-11-09 18:25:00"	"Nov 09, 2026 06:25 PM"	"Nov 09, 2026 06:25 PM"	**Pass**
UT_34	Empty / null date	None	Input: "" and null	Em dash placeholder	Em dash placeholder	**Pass**

is_logged_in()

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
UT_35	No session data	Empty session	Call is_logged_in()	false	false	**Pass**
UT_36	user_id and role set	Session has user_id = 1, role = citizen	Call is_logged_in()	true	true	**Pass**
UT_37	Only user_id set	Session has user_id = 1, no role	Call is_logged_in()	false	false	**Pass**

format_case_id()

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
UT_38	Standard Case ID	Function copied from the page where it is defined	report_id = 14, submitted_at = "2026-11-09 10:00:00"	"CR-2026-00014"	"CR-2026-00014"	**Pass**
UT_39	Year taken from submission date	Same	report_id = 7, submitted_at = "2025-01-05 08:30:00"	"CR-2025-00007"	"CR-2025-00007"	**Pass**
UT_40	ID longer than 5 digits	Same	report_id = 123456, submitted_at = "2026-03-01 09:00:00"	"CR-2026-123456" (no truncation)	"CR-2026-123456"	**Pass**

6. Integration Test Cases

Key end-to-end flows and page-level behaviour across the Citizen, Police and Admin modules.

Login

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_01	Valid admin login	User is on the Login page; seeded admin exists	1. Enter admin@crms.local 2. Enter Admin@123 3. Click Login	Redirected to Admin Dashboard with message "Welcome back, System Administrator!"	Redirected to Admin Dashboard with welcome message	**Pass**
IT_02	Valid citizen login	Registered citizen account exists	1. Enter citizen email 2. Enter correct password 3. Click Login	Redirected to Citizen Dashboard	Redirected to Citizen Dashboard	**Pass**
IT_03	Valid police login	Police account created by admin	1. Enter police email 2. Enter correct password 3. Click Login	Redirected to Police Dashboard	Redirected to Police Dashboard	**Pass**

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_04	Wrong password	User is on the Login page	1. Enter valid email 2. Enter wrong password 3. Click Login	Error: "Invalid email or password."	Error "Invalid email or password." displayed	**Pass**
IT_05	Unregistered email	User is on the Login page	1. Enter unknown@example.com 2. Enter any password 3. Click Login	Same generic error (no account disclosure)	Same generic error displayed	**Pass**
IT_06	Both fields empty	User is on the Login page	1. Leave email and password blank 2. Click Login	Error: "Please enter both your email and password."	Error displayed	**Pass**
IT_07	Password empty	User is on the Login page	1. Enter valid email 2. Leave password blank 3. Click Login	Error: "Please enter both your email and password."	Error displayed	**Pass**
IT_08	SQL injection check	User is on the Login page	1. Enter ' OR 1=1 -- as email 2. Enter any password 3. Click Login	Access denied; no database error	Error "Invalid email or password."; no access (prepared statement)	**Pass**
IT_09	Invalid CSRF token	Login form opened; token altered with browser tools	1. Change hidden csrf_token value 2. Submit valid credentials	Error: "Invalid form submission. Please refresh the page and try again."	Login blocked; CSRF error displayed	**Pass**
IT_10	Forgot password link	User is on the Login page	1. Click "Forgot password?"	Password reset page opens	HTTP 404 Not Found (forgot_password.php does not exist)	**Fail**
IT_11	Logged-in user opens login page	User already logged in	1. Open login.php directly	Redirected to the user's own dashboard	Redirected to the user's own dashboard	**Pass**
IT_12	Logout	User logged in	1. Click Logout 2. Open citizen/dashboard.php	Message "You have been logged out successfully."; dashboard now redirects to login	Session destroyed; redirected to login with "Please log in to continue."	**Pass**

Registration

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_13	Valid registration	User is on the Register page	1. Fill valid name, email, password (8+ chars), matching confirm password 2. Click Register	Account created with role citizen; redirected to Login with success message	Account created as citizen; success message displayed	**Pass**
IT_14	Name too short (boundary 2)	User is on the Register page	Full name = "Al", other fields valid	Error: "Full name must be at least 3 characters long."	Error displayed	**Pass**
IT_15	Name at minimum (boundary 3)	User is on the Register page	Full name = "Ali", other fields valid	Registration accepted	Registration accepted	**Pass**
IT_16	Invalid email format	User is on the Register page	Email = "abc@"	Error: "Please enter a valid email address."	Error displayed	**Pass**
IT_17	Password below minimum (7)	User is on the Register page	Password = "Pass123" (7 characters)	Error: "Password must be at least 8 characters long."	Error displayed	**Pass**
IT_18	Password at minimum (8)	User is on the Register page	Password = "Pass1234" (8 characters)	Registration accepted	Registration accepted	**Pass**
IT_19	Passwords do not match	User is on the Register page	Password and Confirm Password differ	Error: "Passwords do not match."	Error displayed	**Pass**

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_20	Duplicate email	Email already registered	Register again with the same email	Error: "An account with this email already exists. Please log in instead."	Error displayed	**Pass**
IT_21	Phone below minimum length (5)	User is on the Register page	Phone = "12345"	Error: "Please enter a valid phone number."	Error displayed	**Pass**
IT_22	Phone at minimum length (6)	User is on the Register page	Phone = "123456"	Registration accepted	Registration accepted	**Pass**
IT_23	Name above database limit (101 characters)	User is on the Register page	Full name = 101 characters, other fields valid	Validation error about maximum name length	Unhandled database error (Data too long for column full_name); blank error page	**Fail**
IT_24	Email above database limit (101 characters)	User is on the Register page	Valid-format email of 101 characters	Validation error about maximum email length	Unhandled database error (Data too long for column email); blank error page	**Fail**
IT_25	Script tag in name	User is on the Register page	Full name = <script>alert(1)</script>; register then log in	Script is not executed anywhere on the page	Value shown as plain text on the welcome message (output escaped)	**Pass**
IT_26	Role escalation attempt	User is on the Register page	Add role=admin to the POST data using browser tools	Account is still created as citizen	Role is fixed to citizen in the INSERT query	**Pass**

Report Submission

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_27	Valid report without evidence	Citizen logged in	Fill all fields with valid data; no file; click Submit	Report saved with status Pending; notification created; redirected to report details	Report saved as Pending; notification created	**Pass**
IT_28	Valid report with JPG evidence	Citizen logged in	Fill valid data; attach one JPG under 5 MB; Submit	Report and evidence row saved; file stored under uploads/evidence	Report and evidence saved	**Pass**
IT_29	Title below minimum (4)	Citizen logged in	Title = "Thef"	Error: "Title must be between 5 and 150 characters."	Error displayed	**Pass**
IT_30	Title at minimum (5)	Citizen logged in	Title = "Theft"	Report accepted	Report accepted	**Pass**
IT_31	Title above maximum (151)	Citizen logged in	Title = 151 characters	Error: "Title must be between 5 and 150 characters."	Error displayed	**Pass**
IT_32	Description below minimum (19)	Citizen logged in	Description = 19 characters	Error: "Description must be at least 20 characters..."	Error displayed	**Pass**
IT_33	Description at minimum (20)	Citizen logged in	Description = 20 characters	Report accepted	Report accepted	**Pass**
IT_34	Future incident date	Citizen logged in	Incident date = tomorrow	Error: "Incident date cannot be in the future."	Error displayed	**Pass**
IT_35	Incident date = today (boundary)	Citizen logged in	Incident date = today	Report accepted	Report accepted	**Pass**
IT_36	Impossible date	Citizen logged in	Incident date = 2026-02-30 (typed or edited in request)	Error: "Please enter a valid incident date."	Error displayed	**Pass**

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_37	Invalid category	Citizen logged in	Change category_id to 9999 using browser tools	Error: "Please select a valid crime category."	Error displayed	**Pass**
IT_38	Five evidence files (boundary)	Citizen logged in	Attach 5 valid files; Submit	Report and all 5 files saved	Report and 5 files saved	**Pass**
IT_39	Six evidence files	Citizen logged in	Attach 6 valid files; Submit	Error: "You can upload up to 5 evidence files per report."	Error displayed; nothing saved	**Pass**
IT_40	Invalid evidence type with valid fields	Citizen logged in	Attach virus.exe; other fields valid; Submit	Report saved; warning that the file was not uploaded; file not stored	Report saved; warning displayed; file not stored	**Pass**

Police Verification

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_41	Verify a pending report	Police logged in; report is Pending	Open Verify page; choose Verified; remarks "Evidence is sufficient"; Submit	Status = Verified; verified_by and verified_at set; citizen notified	Status updated; citizen notification created	**Pass**
IT_42	Reject a pending report	Police logged in; report is Pending	Choose Rejected; enter remarks; Submit	Status = Rejected; citizen notified with rejection message	Status updated; notification created	**Pass**
IT_43	Remarks below minimum (4)	Police logged in; report is Pending	Remarks = "Okay"	Error: "...at least 5 characters"	Error displayed	**Pass**
IT_44	No decision selected	Police logged in; report is Pending	Leave decision empty; enter remarks; Submit	Error: "Please choose whether to verify or reject this report."	Error displayed	**Pass**
IT_45	Verify an already verified report	Report status is Verified	Open verify_report.php?id=<report id>	Warning; redirected to report details; no change	Warning displayed; redirected	**Pass**
IT_46	Invalid report id	Police logged in	Open verify_report.php?id=abc	Error: "Invalid report reference."	Error displayed; redirected to report list	**Pass**
IT_47	Report id does not exist	Police logged in	Open verify_report.php?id=99999	Error: "That report was not found."	Error displayed; redirected to report list	**Pass**

Investigation

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_48	Add Investigating update	Police logged in; report is Verified	Select Investigating; remarks "Suspect identified"; Submit	Investigation row inserted; report status = Investigating; citizen notified	Row inserted; status synced; notification created	**Pass**
IT_49	Mark case as Resolved	Police logged in; report is Investigating	Select Resolved; remarks "Case solved"; Submit	Report status = Resolved; investigation row inserted; citizen notified	Status changed to Resolved; row inserted; notification created	**Pass**
IT_50	Close a resolved case	Police logged in; report is Resolved	Select Closed (finalize and officially close case file); remarks "File archived"; Submit	Report status = Closed; investigation row inserted; citizen notified	Status changed to Closed; row inserted; notification created	**Pass**

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_51	Invalid status value	Police logged in; report is Verified	Change the status value to "Verified" using browser tools; Submit	Error: "Please select a valid investigation status."	Error displayed	**Pass**
IT_52	Investigation remarks below minimum (4)	Police logged in; report is Verified	Select Investigating; remarks = "Done"	Error: "Please provide investigation remarks (at least 5 characters)."	Error displayed	**Pass**
IT_53	Investigation remarks at minimum (5)	Police logged in; report is Verified	Select Investigating; remarks = "Done."	Update accepted	Update accepted	**Pass**
IT_54	Investigation on pending report	Report status is Pending	Open investigation.php?id=<report id>	Warning: report must be verified first	Warning displayed; redirected	**Pass**
IT_55	Resolved case can still be updated	Report status is Resolved	Open investigation.php?id=<report id>	Form opens so the officer can close the case	Form opens; Investigating, Resolved and Closed are offered	**Pass**
IT_56	Investigation page for a closed case	Report status is Closed	Open investigation.php?id=<report id>	Warning: "This case file is officially closed and archived."; redirected to report details	Warning displayed; redirected	**Pass**

Case Status History

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_57	History recorded when a report is verified	Police logged in; report is Pending; case_status_history table exists	1. Verify the report with remarks 2. Run SELECT * FROM case_status_history WHERE report_id = <id>	One row: Pending to Verified, with remarks and officer id	No row found (create_case_status_history() is never called)	**Fail**
IT_58	History recorded on investigation update	Report is Verified	1. Add an Investigating update 2. Query case_status_history for the report	One row: Verified to Investigating	No row found (create_case_status_history() is never called)	**Fail**
IT_59	History table in database script	Fresh import of database.sql	Import database.sql; run DESCRIBE case_status_history	Table created with foreign keys to crime_reports (CASCADE) and users (SET NULL)	Table created with the expected columns and foreign keys	**Pass**

Dashboards and Case Tracking

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_60	Citizen dashboard counts Resolved and Closed together	Citizen has one Resolved and one Closed report	Open citizen/dashboard.php	Resolved card shows 2	Resolved card shows 2	**Pass**
IT_61	Admin dashboard counts Resolved and Closed together	Admin logged in; one Resolved and one Closed report	Open admin/dashboard.php	Resolved/Closed card shows 2	Card shows 2	**Pass**
IT_62	Admin dashboard with no reports	Admin logged in; no reports in database	Open admin/dashboard.php	Page loads; all counts show 0; charts render with zero values	Page loads; counts show 0; charts render with zero values	**Pass**

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_63	Police dashboard Resolved card counts Closed cases	Police logged in; one Resolved and one Closed report	Open police/dashboard.php	"Resolved System-Wide" card shows 2, matching the lifecycle breakdown	Card shows 1 (Closed not counted) while the Resolved / Closed line in the lifecycle breakdown shows 2	**Fail**
IT_64	Police dashboard with no reports	Police logged in; no reports in database	Open police/dashboard.php	Page loads; all percentages show 0%; no division error	Page loads; percentages show 0% (denominator forced to 1)	**Pass**
IT_65	Police lifecycle percentages	One Pending, one Verified and one Investigating report	Open police/dashboard.php	Each of the three statuses shows 33%	Each shows 33% (rounded)	**Pass**
IT_66	Admin dashboard charts	Admin logged in; reports in several statuses and categories; internet available	Open admin/dashboard.php	Doughnut chart (status) and bar chart (category) show the same counts as the page figures	Both charts drawn from the same counts	**Pass**
IT_67	Citizen timeline for a closed case	Citizen owns a Closed report	Open citizen/report_details.php?id=<id>	Submitted, Verified, Under Investigation, Resolved and Closed all shown as completed	All five steps shown as completed	**Pass**
IT_68	Citizen timeline for a rejected report	Citizen owns a Rejected report	Open citizen/report_details.php?id=<id>	Timeline shows Submitted and Rejected only	Submitted and Rejected steps shown	**Pass**
IT_69	Case ID shown on pages	Report with id 14 submitted in 2026	Open the verify or investigation page	Badge shows CR-2026-00014	Badge shows CR-2026-00014	**Pass**
IT_70	Search by Case ID	Police logged in; report id 3 exists	Search reports for "CR-2026-00003"	Report 3 appears in the results	Report 3 listed (id is extracted from the Case ID)	**Pass**
IT_71	Search by Case ID in lower case	Police logged in; report id 3 exists	Search reports for "cr-2026-00003"	Report 3 appears in the results	Report 3 listed (pattern is case-insensitive)	**Pass**
IT_72	Filter by Closed status	Police logged in	Choose status Closed in the filter	Only Closed reports are listed	Only Closed reports listed (Closed is a valid filter value)	**Pass**

Filters and Report Details

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_73	Police filter: last 7 days	Police logged in; reports exist from different dates	Choose Last 7 Days; click Search	Only reports submitted in the last 7 days	Only reports from the last 7 days listed	**Pass**
IT_74	Police filter: last 30 days	Same	Choose Last 30 Days; click Search	Only reports submitted in the last 30 days	Only reports from the last 30 days listed	**Pass**
IT_75	Police filter: this month	Same	Choose This Month; click Search	Only reports submitted in the current month and year	Only current month reports listed	**Pass**

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_76	Police filter: exact date	Same	Choose Specific Exact Date; pick a date; Search	Only reports filed on that calendar date	Only reports filed on that date listed	**Pass**
IT_77	Exact date selected but no date entered	Police logged in	Choose Specific Exact Date; leave the date empty; Search	All reports shown (no date condition)	All reports listed	**Pass**
IT_78	Invalid exact date in the URL	Police logged in	Open police/reports.php?date_range=exact&exact_date=abc	No error; no report matches	Page loads; empty result list	**Pass**
IT_79	Admin filter: last 30 days	Admin logged in	Open admin/reports.php; choose Last 30 Days; Search	Only reports from the last 30 days	Only reports from the last 30 days listed	**Pass**
IT_80	Clear active filters	A filter is active	Click the clear button	All reports shown; filters reset	All reports listed; clear button shown only while a filter is active	**Pass**
IT_81	Admin report details page	Admin logged in; report exists	Open admin/reports.php; click View on a report	Page shows Case ID, category, location, description, evidence, investigation updates and citizen contact	Page shows all sections	**Pass**
IT_82	Admin details with invalid id	Admin logged in	Open admin/report_details.php?id=0	Error: "Invalid report reference."; redirected to report list	Error displayed; redirected	**Pass**
IT_83	Admin details with unknown id	Admin logged in	Open admin/report_details.php?id=99999	Error: "That report was not found."; redirected	Error displayed; redirected	**Pass**
IT_84	Script in description on admin details page	Report description contains <script>alert(1)</script>	Open admin/report_details.php?id=<id>	Script shown as text and not executed	Output escaped; script not executed	**Pass**
IT_85	Print / Export PDF button	Any report details page (citizen, police or admin)	Click Print / Export PDF	Print dialog opens; navigation bar and buttons hidden in the print view	Print dialog opens; navigation and buttons hidden by the print stylesheet	**Pass**

Access Control

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_86	Unauthenticated access	User not logged in	Open citizen/dashboard.php directly	Redirected to login with "Please log in to continue."	Redirected to login with warning	**Pass**
IT_87	Citizen opens admin page	Citizen logged in	Open admin/dashboard.php	Message "You are not authorized to access that page."; redirected to citizen dashboard	Message displayed; redirected	**Pass**
IT_88	Police opens citizen submit page	Police logged in	Open citizen/submit_report.php	Not authorized; redirected to police dashboard	Not authorized; redirected	**Pass**
IT_89	Citizen views another citizen's report	Two citizens exist; Citizen A logged in	Open citizen/report_details.php?id=<Citizen B report id>	Message "not found or does not belong to your account"	Message displayed; redirected to report history	**Pass**
IT_90	Police opens admin report details	Police logged in	Open admin/report_details.php?id=<report id>	Not authorized; redirected to police dashboard	Not authorized; redirected	**Pass**

Admin Management

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Actual Result	Status
IT_91	Category name below minimum (2)	Admin logged in	Create category "AB"	Error: "Category name must be between 3 and 100 characters."	Error displayed	**Pass**
IT_92	Duplicate category name	Category "Theft" exists	Create category "Theft"	Error: "A category with this name already exists."	Error displayed	**Pass**
IT_93	Delete category in use	Category has at least one report	Click Delete on that category	Deletion blocked with explanatory message	Message: "cannot be deleted because one or more crime reports are already assigned to it."	**Pass**
IT_94	Delete unused category	Category has no reports	Click Delete and confirm	Category deleted	Category deleted	**Pass**
IT_95	Delete officer with investigations	Officer has investigation records	Click Delete on the officer	Deletion blocked with explanatory message	Message: "cannot be deleted because they have existing investigation records..."	**Pass**
IT_96	Search with special characters	Police logged in	Search reports for: '; DROP TABLE users; --	No SQL error; only matching reports (none) shown	No error; empty result list	**Pass**

7. Test Execution Summary

Unit tests

Function	Total	Pass	Fail
sanitize_input()	6	5	1
validate_file_upload()	14	14	0
generate_unique_filename()	3	3	0
get_status_badge_class()	4	4	0
CSRF token functions	5	5	0

Function	Total	Pass	Fail
format_date()	2	2	0
is_logged_in()	3	3	0
format_case_id()	3	3	0
Total	40	39	1

Integration tests

Module	Total	Pass	Fail
Login	12	11	1
Registration	14	12	2
Report Submission	14	14	0
Police Verification	7	7	0
Investigation	9	9	0
Case Status History	3	1	2
Dashboards and Case Tracking	13	12	1
Filters and Report Details	13	13	0
Access Control	5	5	0
Admin Management	6	6	0
Total	96	90	6

Overall (before fixes)

Suite	Total	Pass	Fail
Unit tests	40	39	1
Integration tests	96	90	6
All tests	136	129	7

Pass rate before fixes: 94.9%. After the fixes in Section 9 were applied, all 7 failed cases were re-run and passed, giving 136 of 136 passed.

8. Coverage Summary

Coverage is reported as function coverage and feature (requirement) coverage.

Area	Items	Covered by	Coverage
Helper functions (functions.php)	13 functions. Tested directly: sanitize_input, validate_file_upload, generate_unique_filename, get_status_badge_class, generate_csrf_token, verify_csrf_token, format_date. Exercised through page flows: redirect, set_flash_message, get_flash_message, create_notification, create_case_status_history. Not called from any page: get_case_status_history	UT_01 to UT_34 and the integration tests	12 of 13 functions (92%)
Case ID helper (format_case_id)	Defined inside the pages, so it was tested as a copy of the function	UT_38 to UT_40, IT_69	100%
Authentication functions (auth.php)	is_logged_in, login_user, logout_user, require_login, require_role, redirect_to_dashboard	UT_35 to UT_37, IT_01 to IT_12, IT_86 to IT_90	100%
Citizen features	Register, login, submit report, evidence upload, view own report, case timeline	IT_13 to IT_40, IT_67 to IT_68	Core flows covered

Area	Items	Covered by	Coverage
Police features	Verify or reject, investigation updates (Investigating, Resolved, Closed), status history, report search and Case ID search	IT_41 to IT_59, IT_70 to IT_72	Core flows covered
Dashboards	Citizen, police and admin status counters	IT_60 to IT_72	All three dashboards covered
Search, filters and details pages	Date filters (7 days, 30 days, month, exact date), Case ID search, admin report details, print view	IT_73 to IT_85, IT_70 to IT_72	All filters and the details page covered
Admin features	Categories, citizens, officers (create validation, delete rules)	IT_91 to IT_96	Category rules and officer deletion rules covered
Security controls	CSRF, SQL injection, XSS escaping, role checks, ownership checks, upload validation	UT_03, UT_15, UT_30 to UT_32, IT_08, IT_09, IT_25, IT_26, IT_86 to IT_90	Covered

9. Defects Found and Fixed

Defect ID	Description	Found By	Severity	Location	Fix Applied	Retest Result
D_01	Broken link: "Forgot password?" on the login page points to forgot_password.php, which does not exist (HTTP 404).	IT_10	Medium	login.php	Removed the link from the login form. Password reset is outside the approved scope; the reset_token columns remain available for a later feature.	Pass
D_02	Registration crashes on over-long input. A name or email longer than 100 characters passes validation but fails in MySQL (Data too long), giving an unhandled exception and a blank page.	IT_23, IT_24	Medium	register.php, admin/police.php, admin/users.php	Added validation: full name 3 to 100 characters and email at most 100 characters, with clear error messages, in all three forms.	Pass
D_03	Case status history is never recorded. The case_status_history table and create_case_status_history() exist, but no page calls the function, so verification and investigation changes leave no history.	IT_57, IT_58	Medium	police/verify_report.php, police/investigation.php	Added a call to create_case_status_history() after a successful verify or reject decision and after each investigation update, passing the previous status, new status, remarks and officer id.	Pass
D_04	Police dashboard shows different totals on one page. The "Resolved System-Wide" card counts only Resolved cases, while the lifecycle breakdown on the same page counts Resolved and Closed together.	IT_63	Low	police/dashboard.php	The card now shows $totalResolvedOrClosed, the same value the lifecycle breakdown already uses.	Pass
D_05	sanitize_input() removes backslashes from valid text because it calls stripslashes(). Input such as C:Users becomes C:Users when shown back to the user.	UT_05	Low	includes/functions.php	Removed the stripslashes() call (only needed for magic quotes, which no longer exist). trim() and htmlspecialchars() are kept.	Pass

10. Defect Categories Not Covered by the Current Tests

The following categories were identified as gaps in the current suite.

[1] Brute-force protection: login has no attempt limit or lockout, so repeated password guessing is not blocked.

[2] Session handling: no automatic session timeout or idle logout was tested.

[3] Status transition rules: an officer can move a Resolved case back to Investigating. No rule is specified for this, so no test asserts it.

[4] Consistency of multi-step updates: the status update and the investigation row are saved as separate queries without a transaction, so a failure between them was not tested.

[5] Concurrency: two officers acting on the same report at the same time (the code guards this with a status condition in the UPDATE, but it was not load-tested).

[6] Offline use: Chart.js is loaded from a CDN on the dashboards, so the charts were not tested without an internet connection.

[7] Performance: behaviour with a large number of reports, search results and notifications.

[8] Upload security: files are checked by extension and MIME type, but no antivirus or image re-encoding test was done.

[9] Usability and compatibility: responsive layout and different browsers were not covered.

11. Conclusion

A total of 136 test cases (40 unit and 96 integration) were designed using equivalence partitioning and boundary value analysis. 7 test cases failed initially and exposed 5 defects. All defects were fixed and the failed cases were re-run until the whole suite passed.